Deriving the Relationship Between Wheel Angular Speeds, Steering Angle, and Motor Speed

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1. Forward Velocity from Wheel Angular Speeds

Let the rear wheels have angular velocities ω_L and ω_R, and let R be the radius of the wheels. The linear velocity of each wheel is:

$$v_L = R\omega_L, \quad v_R = R\omega_R$$

The vehicle's forward velocity v_x (at the midpoint of the rear axle) is the average of the two:

$$v_x = \frac{v_L + v_R}{2} = \frac{R}{2}(\omega_L + \omega_R) \tag{1}$$

2. Yaw Rate from Wheel Velocities

The yaw rate $\dot{\theta}_z$, i.e., the angular velocity about the vertical axis, can be expressed from the difference in wheel velocities:

$$\dot{\theta}_z = \frac{v_R - v_L}{d_{\text{track}}} = \frac{R(\omega_R - \omega_L)}{d_{\text{track}}} \tag{2}$$

3. Yaw Rate from Steering Geometry

Alternatively, using the kinematic bicycle model, the yaw rate is:

$$\dot{\theta}_z = \frac{v_x \cdot \tan(\delta_f)}{d_{\text{wheelbase}}} \tag{3}$$

Substituting Equation (1) into (3):

$$\dot{\theta}_z = \frac{\frac{R}{2}(\omega_L + \omega_R) \cdot \tan(\delta_f)}{d_{\text{wheelbase}}} = \frac{R(\omega_L + \omega_R) \cdot \tan(\delta_f)}{2d_{\text{wheelbase}}} \tag{4}$$

4. Equating Both Yaw Expressions

Equating equations (2) and (4):

$$\frac{R(\omega_R - \omega_L)}{d_{\text{track}}} = \frac{R(\omega_L + \omega_R) \cdot \tan(\delta_f)}{2d_{\text{wheelbase}}}$$

Cancel R and rearrange:

$$\frac{\omega_R - \omega_L}{d_{\text{track}}} = \frac{(\omega_L + \omega_R) \cdot \tan(\delta_f)}{2d_{\text{wheelbase}}} \tag{5}$$

5. Solving for the Ratio ω_R/ω_L

Multiplying both sides by $2d_{\text{wheelbase}} \cdot d_{\text{track}}$:

$$2d_{\text{wheelbase}}(\omega_R - \omega_L) = d_{\text{track}}(\omega_L + \omega_R)\tan(\delta_f)$$

Rearranging terms:

$$\omega_R(2d_{\text{wheelbase}} - d_{\text{track}}\tan(\delta_f)) = \omega_L(2d_{\text{wheelbase}} + d_{\text{track}}\tan(\delta_f))$$

Solving for the ratio:

$$\frac{\omega_R}{\omega_L} = \frac{2d_{\text{wheelbase}} + d_{\text{track}}\tan(\delta_f)}{2d_{\text{wheelbase}} - d_{\text{track}}\tan(\delta_f)} \tag{6}$$

Normalizing by dividing numerator and denominator by d_{track}:

$$\frac{\omega_R}{\omega_L} = \frac{k + \tan(\delta_f)}{k - \tan(\delta_f)}, \tag{7}$$

$$\text{where } k = \frac{2d_{\text{wheelbase}}}{d_{\text{track}}} \tag{8}$$

6. Using Motor Speed Constraint

Let the motor driving both wheels through a differential have speed ω_1, and gear ratio i_{12}. Then:

$$\omega_L + \omega_R = \frac{\omega_1}{i_{12}} \tag{9}$$

Let $\rho = \frac{\omega_R}{\omega_L}$. Then:

$$\omega_R = \rho\omega_L \Rightarrow \omega_L(1+\rho) = \frac{\omega_1}{i_{12}} \Rightarrow \omega_L = \frac{\omega_1}{i_{12}(1+\rho)}, \quad \omega_R = \frac{\rho \cdot \omega_1}{i_{12}(1+\rho)}$$

7. Final Expressions

Substituting $\rho = \frac{k+\tan(\delta_f)}{k-\tan(\delta_f)}$, we get:

$$\boxed{\omega_L = \frac{1}{1+\rho} \cdot \frac{\omega_1}{i_{12}}, \quad \omega_R = \frac{\rho}{1+\rho} \cdot \frac{\omega_1}{i_{12}}}$$

These equations express the left and right rear wheel angular speeds directly in terms of the steering angle δ_f, vehicle geometry, and motor speed.